Rui Zhang

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EDUCATION

Ocean University of China

M.Eng. in Environmental Geological Engineering Sep 2023 – Jun 2026

• Overall GPA: 3.76 / 4.0

• Research focus: Earthquake-triggered submarine landslides and marine geohazards

• Supervisor: Prof Dong Wang

Ocean University of China

B.Eng. in Environmental Engineering Sep 2019 – Jun 2023

• Overall GPA: 3.77 / 4.0

PUBLICATIONS

• **Zhang, R.**, Wang, D., Zhang, W. (2026). *A depth-averaged method modeling three-dimensional submarine landslides caused by earthquakes*. *Computers and Geotechnics*. Accepted for publication

RESEARCH EXPERIENCE

National Science Foundation for Distinguished Young Scholars Project

Mechanisms of Shallow Submarine Landslides in the Northern South China

• Developed a new depth-averaged numerical model capable of simulating the complete co-seismic evolution of submarine landslides, incorporating three-component seismic loading, strain-softening behavior, and terrain curvature effects.

• Developed a GPU-accelerated 3D landslide simulation framework (~10× faster than CPU).

CNOOC Industry Collaboration Project

Evolution and Mechanism of Submarine Landslides in a Continental Slope Region of the South China Sea

• Developed a predictive model to estimate the maximum impact velocity, run-out distance, final deposition extent, and thickness of submarine landslide gravity flows.

• Conducted fully dynamic simulations of the entire landslide process under varying earthquake intensities, generating four-dimensional (3D space + time) visualizations of its evolution.

SELECTED AWARDS AND HONORS

• National Scholarship of China, Ministry of Education of China (2021 & 2022)

• Outstanding Graduate of Shandong Province (Class of 2023)

• Honors Bachelor's Degree, Ocean University of China (2023)

• First-Class Scholarship for Graduates, Ocean University of China (2024)

• First-Class Scholarship for Undergraduates, Ocean University of China (2020 & 2021 &2022)

TECHNICAL SKILLS

• Programming skills: MATLAB (5-years experience; numerical modeling, GPU acceleration, visualization), Python, Linux (Spectral-element simulations of submarine seismic wave propagation).

• Language: Mandarin (native), English (fluent; IELTS 7.5), German (beginner)